

“JEFFREY: WiFi-Controlled Rocker-Bogie Surveillance Rover ”

A MINOR PROJECT REPORT

*Submitted in
partial fulfillment of the requirement for the award of the degree of*

BACHELOR OF TECHNOLOGY IN MECHATRONICS ENGINEERING



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(Declared Autonomous by Government of Madhya Pradesh and RGPV, Bhopal)**

SESSION: 2022-2026

DEPARTMENT OF MECHATRONICS ENGINEERING



DECLARATION

We hereby declare that the work, which is being presented in this Minor Project Report, entitled **JEFFERY: WiFi-Controlled Rocker-Bogie Surveillance Rover** in partial fulfillment for the award of **Bachelor of Technology** degree in **Mechatronics Engineering**, submitted in the Department of Mechatronics Engineering, Jabalpur Engineering College, Jabalpur, is an authentic record of our own work carried out under the guidance of **Prof. Shudhanshu Dole, Mechatronics Engineering, Jabalpur Engineering College, Jabalpur.**

We have not submitted the matter embodied in this report for award of any other degree or diploma.

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CERTIFICATE

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CERTIFICATE OF APPROVAL

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INTERNAL EXAMINER

EXTERNAL EXAMINER

DATE:

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ABSTRACT

JEFFREY is a WiFi-controlled six-wheeled surveillance rover built on NASA's Rocker-Bogie suspension system. The rover is designed to navigate rough, uneven, and obstacle-laden terrain that conventional wheeled robots fail on. The mechanical chassis is constructed from PVC pipes and ACP sheet, employing a passive differential Rocker-Bogie linkage that keeps all six wheels in constant contact with the ground without any springs or active actuators. The electronics system comprises an Arduino WiFi board as the main controller, three L298N dual H-bridge motor drivers, six 12V DC geared motors operating via skid-steer drive, and an ESP32-CAM module for live video streaming. Control commands are sent wirelessly from a smartphone or laptop browser over WiFi, providing real-time operator feedback. The prototype was successfully tested on paved concrete, tiled surfaces, and platform steps, achieving chassis tilt variance as low as $\pm 0.07^\circ$, body stability of 0.050σ , and successful obstacle clearance. This report documents the design, fabrication, circuit implementation, testing results, and future enhancements planned for the system.

Keywords: *Rocker-Bogie Suspension, Surveillance Rover, Arduino, ESP32-CAM, WiFi Control, Skid-Steer, L298N Motor Driver, Terrain Navigation*

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CHAPTER 1 – INTRODUCTION

1.1 Introduction

JEFFREY (Joint Engineering Framework for Field Recon and Exploration Yielder) is a WiFi-controlled six-wheeled surveillance rover built on NASA's Rocker-Bogie suspension system. The rover is designed to navigate rough, uneven, and obstacle-laden terrain while providing live video feedback to a remote operator. It is constructed from PVC pipes and an ACP (Aluminium Composite Panel) base plate, controlled by an Arduino WiFi board, driven by three L298N motor driver modules, and equipped with an ESP32-CAM module for real-time video streaming.

Surveillance in remote, hazardous, or inaccessible areas is a growing need across sectors such as border security, disaster response, wildlife monitoring, and industrial inspection. Conventional wheeled robots fail on challenging terrains due to rigid chassis designs that cannot adapt to uneven surfaces. The Rocker-Bogie suspension, originally developed by NASA's Jet Propulsion Laboratory (JPL) and proven on Mars rovers including Curiosity (2012) and Perseverance (2021), solves this by using a passive linkage that keeps all six wheels in constant ground contact — without springs or active actuators. This project adapts that proven mechanism into a low-cost terrestrial prototype.

1.2 Literature Review

Harrington and Voorhees (2004) documented the mechanical challenges of designing the Rocker-Bogie suspension for the Mars Exploration Rover program, establishing the theoretical basis for passive load distribution across six wheels on uneven terrain [1]. Singh and Jain (2020) demonstrated a scaled planetary rover prototype using the Rocker-Bogie principle with DC motors and validated it on indoor test surfaces, confirming the feasibility of the design for student-level projects [2]. Choudhury et al. (2023) conducted a comparative analysis of rocker-bogie suspension performance versus conventional rigid-chassis robots, concluding that the passive differential mechanism provides superior terrain adaptability at minimal added mechanical complexity [3]. Studies on Arduino-based WiFi control systems (Last Minute Engineers, 2026; Arduino Project Hub, 2026) have confirmed the reliability of L298N motor drivers with Arduino for multi-motor skid-steer configurations [4][5]. The ESP32-CAM's suitability for low-latency MJPEG surveillance streaming in embedded robotics applications has been documented in embedded systems literature (Espressif, 2024)

[9].

1.3 Objectives

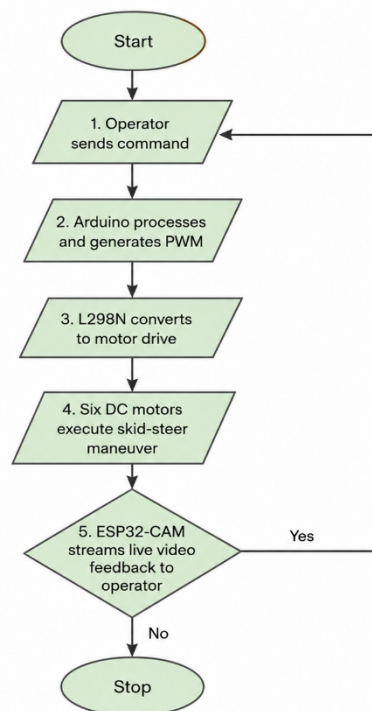
The primary objectives of the JEFFREY project are:

- To design and fabricate a six-wheeled Rocker-Bogie mechanical chassis using PVC pipes and ACP sheet capable of navigating surfaces with tilt angles up to 25°.
- To implement a three-L298N motor driver circuit controlled by an Arduino WiFi board for independent left and right side motor control (skid-steer drive).
- To integrate an ESP32-CAM module providing a live MJPEG video stream accessible from any WiFi-connected browser.
- To develop a WiFi web-server control interface on the Arduino allowing directional commands (Forward, Backward, Left, Right, Stop) from a smartphone or laptop.
- To test and validate the complete system on at least three distinct terrain types and document quantitative performance metrics.

CHAPTER 2 – WORKING PRINCIPLE

2.1 System Overview

JEFFREY operates as a closed-loop tele-operated system. The operator issues motion commands from a WiFi-connected device (smartphone or laptop browser). The Arduino WiFi board hosts a web server that receives these HTTP requests, decodes the intended direction, and generates the appropriate HIGH/LOW logic signals on its digital I/O pins. These signals are fed to three L298N motor driver modules that amplify them into sufficient current and voltage to drive six 12V DC geared motors. Simultaneously, the ESP32-CAM module — connected to the Arduino via serial (TX/RX) for synchronization — captures live frames from its onboard OV2640 camera sensor and streams them as an MJPEG feed over a separate WiFi connection back to the operator's browser.



2.2 Rocker-Bogie Suspension Mechanics

The Rocker-Bogie is a two-part passive linkage system. The "rocker" is a long rigid link that pivots centrally about the chassis. Its front end carries one wheel; its rear end connects

to the "bogie" sub-link. The bogie itself is a shorter link that carries two wheels — one at each end — and pivots about its connection to the rocker's rear end.

This geometry provides three key mechanical advantages:

- **Load distribution:** Each of the six wheels carries approximately one-sixth of the vehicle weight at all times due to the passive differential that connects the two rocker arms via a differential bar running across the chassis. This prevents any single wheel from losing contact under normal terrain variations.
- **Obstacle climbing:** When the front bogie wheel encounters a vertical obstacle (step, curb, rock), the bogie link tilts, raising the front wheel over the obstacle. The middle and rear wheels then follow sequentially. The rover can climb steps up to approximately $1.5\times$ its wheel diameter without any active control.
- **Chassis leveling:** The differential bar passively transfers tilt from one rocker to the other, keeping the chassis roll angle at approximately half the terrain tilt angle. On a 25° slope, the chassis tilts approximately 12.5° — keeping the center of gravity within the support polygon and maintaining stability.

The absence of springs means there are no resonance frequencies to excite and no spring fatigue over time. The absence of active actuators means zero motor overhead, zero control complexity, and zero failure modes attributable to the suspension itself.

2.3 Electrical Control Flow

The control pathway begins at the operator's WiFi device. A simple web page (hosted by the Arduino) presents directional buttons. Each button press sends an HTTP GET request (e.g., /forward, /left) to the Arduino's IP address. The Arduino's WiFiServer library parses the request and calls the corresponding motor function.

The motor function sets digital output pins D2–D13 HIGH or LOW in specific combinations to drive the L298N IN1–IN4 inputs. Each L298N controls two DC motors (one on the left side, one on the right side of the chassis). Skid-steer turning is achieved by driving the left-side motors forward while the right-side motors run backward (or vice versa), causing the chassis to pivot in place.

2.4 Vision System

The ESP32-CAM module hosts its own web server (using its built-in WiFi stack) on a separate IP address. It initializes the OV2640 camera sensor at a configurable resolution (default: VGA 640×480) and serves an MJPEG stream. The operator opens this stream in any browser tab on the same WiFi network. The stream is effectively real-time (approximately 10–25 fps depending on network load and resolution). Camera parameters such as brightness, contrast, saturation, AWB, and face detection can be adjusted via the ESP32-CAM's built-in settings panel.

CHAPTER 3 – COMPONENTS

3.1 Mechanical Components

Table 3.1 lists the mechanical components used in the construction of the JEFFREY rover chassis.

Table 3.1: Mechanical Components List

S.No.	Component	Specification	Quantity
1	PVC Pipes	18 cm, 15 cm, 13 cm lengths	As required
2	PVC Elbow Joints	Standard 90° elbows	As required
3	PVC T-Joints	Standard T-junction	As required
4	ACP Sheet Base Plate	Aluminium Composite Panel	1 piece
5	Rubber Pneumatic Wheels	10 cm diameter	6 pieces
6	Rocker-Bogie Link Assembly	Passive differential mechanism	1 set (2 sides)
7	Nuts, Bolts & Screws	M3, M4 assorted	As required

3.2 Electronic Components

Table 3.2 lists all electronic components used in the JEFFREY rover.

Table 3.2: Electronic Components List

S.No.	Component	Specification	Quantity
1	Arduino WiFi Board (UNO R4 WiFi)	ATmega32U4 + ESP32-S3, WiFi 2.4 GHz	1
2	ESP32-CAM	OV2640 Camera, WiFi 802.11 b/g/n	1
3	L298N Motor Driver Module	Dual H-Bridge,	3

		2A/channel, 5–35V	
4	12V DC Geared Motor	60 RPM, 12V, Yellow Gear Motor	6
5	Breadboard	Full-size 830 tie-points	1
6	Jumper Wires	Male-Male, Male-Female assorted	As required
7	Buck Converter Module	LM2596 Step-Down, adjustable output	1

3.3 Power Components

- 12V Li-ion Battery Pack (drive system): Powers all three L298N modules and six DC motors.
- 5V Power Bank: Powers the Arduino WiFi board and ESP32-CAM via USB.
- Power Distribution Wiring Harness: Custom wired distribution to route 12V to L298Ns and 5V to logic boards.

3.4 Communication Components

- WiFi (built-in Arduino WiFi board): Primary control channel. Arduino hosts a web server; operator connects browser to Arduino IP.
- ESP32-CAM WiFi stream: Secondary stream for live video. Operates on same WiFi network as Arduino.
- Future – LoRa SX1278 (433 MHz): Planned for long-range alert transmission beyond WiFi range.

CHAPTER 4 – COMPONENT DESCRIPTIONS

4.1 Arduino WiFi Board (UNO R4 WiFi)

The Arduino UNO R4 WiFi serves as the main microcontroller of JEFFREY. It integrates two processors on a single board: an Renesas RA4M1 ARM Cortex-M4 running at 48 MHz with 256 KB flash and 32 KB RAM, and an ESP32-S3 co-processor that handles WiFi 2.4 GHz connectivity. The board is fully backward compatible with the UNO R3 form factor, offering 14 digital I/O pins, 6 analog inputs, hardware UART/SPI/I2C, and a USB-C connector for programming.

In JEFFREY, the Arduino acts as a WiFi web server. It listens for HTTP requests on port 80. Upon receiving a directional command URL (/F, /B, /L, /R, /S), it sets the appropriate motor driver input pins HIGH or LOW. The 14 digital pins are distributed across three L298N motor drivers: pins D2–D7 for L298N-1 (front motors), D8–D11 for L298N-2 (middle motors), and D12–D13 for L298N-3 (rear motors). The board is powered by the 5V power bank via its USB-C port.

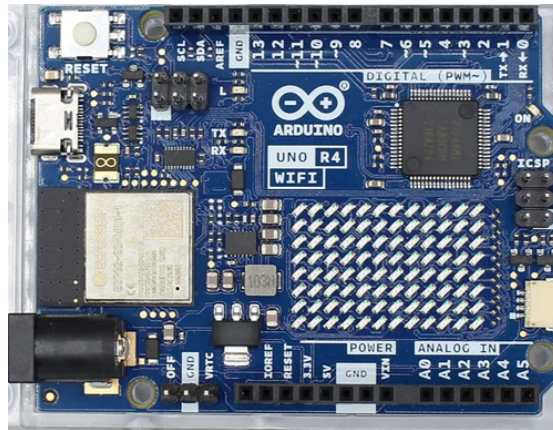


Fig. 4.1: Arduino UNO R4 WiFi (Main Controller)

4.2 ESP32-CAM Module

The ESP32-CAM is a low-cost, small-form-factor development board that combines an Espressif ESP32-S chip with an OV2640 camera module and a microSD card slot. The ESP32-

S features a dual-core Xtensa LX6 processor at up to 240 MHz with 520 KB SRAM. The onboard WiFi (802.11 b/g/n) and Bluetooth stack allow independent wireless connectivity.

The OV2640 camera sensor supports resolutions from 96×96 (QQVGA) up to 1600×1200 (UXGA). In JEFFREY, it operates in VGA (640×480) mode to balance video quality with streaming latency. The module hosts its own CameraWebServer (a publicly available Arduino sketch) that exposes an MJPEG stream at [http://\[ESP32-IP\]/stream](http://[ESP32-IP]/stream) and a full settings panel at [http://\[ESP32-IP\]](http://[ESP32-IP]). The stream is accessible from any browser on the same WiFi network with virtually zero configuration



Fig. 4.2: ESP32-CAM Module with OV2640 Camera

4.3 L298N Dual H-Bridge Motor Driver

The L298N is a monolithic integrated circuit manufactured by STMicroelectronics. It contains two full H-bridges, each capable of driving one DC motor or one winding of a stepper motor. Key electrical specifications are: supply voltage range 5–46V, output current up to 2A per channel (4A peak), and logic voltage 5V TTL compatible. Each H-bridge accepts two logic inputs (IN1/IN2 for one channel, IN3/IN4 for the other) and produces corresponding motor outputs (OUT1/OUT2, OUT3/OUT4).

In JEFFREY, three L298N modules are used — one per pair of motors. Each module drives two DC motors simultaneously: both wheels on one axle (left and right). The enable pins (ENA, ENB) are tied HIGH to allow full-speed operation. Direction is controlled by the IN1–IN4 logic inputs from the Arduino. The onboard 5V regulator on the L298N module (which takes the 12V motor supply and outputs 5V) is deliberately bypassed — Arduino is

powered separately from the power bank — to avoid voltage drop issues under motor load.

Fig. 4.3: L298N Dual H-Bridge Motor Driver Module

4.4 12V DC Geared Motor (60 RPM)

Six identical 12V DC geared motors are used — one per wheel. These are the widely available yellow gear motors (TT motors with gearbox) rated at 60 RPM at 12V no-load. The gearbox reduces the motor shaft speed from ~3000 RPM (bare motor) to 60 RPM output shaft speed with a gear ratio of approximately 48:1. This trades speed for torque, providing significantly higher torque than a direct-drive motor of the same size.

Each motor is rated at approximately 200–250 mA under typical load conditions and up to 1A under stall. The 60 RPM output shaft drives the 10 cm diameter rubber wheels, giving a theoretical top speed of approximately π m/s. The motor is powered by a 12V DC supply at full speed.

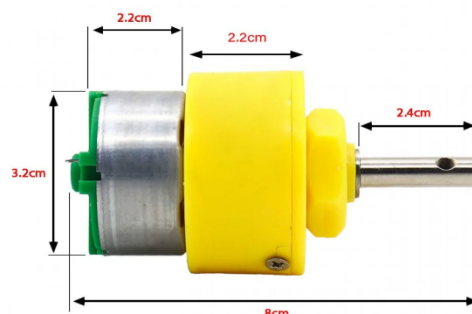


Fig. 4.4: 12V DC Geared



Motor, 60 RPM (Yellow Gear

Motor)

4.5 12V Li-ion Battery Pack

The drive system is powered by a 12V Li-ion battery pack. Li-ion cells provide a high energy density (approximately 150–200 Wh/kg) and a relatively flat discharge curve compared to NiMH or lead-acid alternatives, making them well-suited for mobile robotics where weight and energy density are important. The battery pack used in JEFFREY is a 3S1P configuration (three cells in series): nominal voltage 11.1V, typical maximum charge voltage 12.6V.

The capacity used is approximately 2000–3000 mAh, providing an estimated operational runtime of 45–90 minutes at moderate speed (depending on terrain and motor loading). The battery connects directly to the L298N power input terminals and to the input of the buck converter.



Fig. 4.5: 12V Li-ion Battery Pack (3S Configuration)

4.6 Rubber Pneumatic Wheels (10 cm)

Six rubber pneumatic wheels of 10 cm outer diameter are used. Pneumatic (air-filled) tires provide passive compliance — the air inside deforms slightly under load, absorbing micro-vibrations and small surface irregularities. This supplements the macro-level compliance provided by the Rocker-Bogie suspension and results in smoother video from the ESP32-CAM. The rubber material provides adequate traction on dry, wet, and muddy surfaces due to its high coefficient of friction against most ground materials.

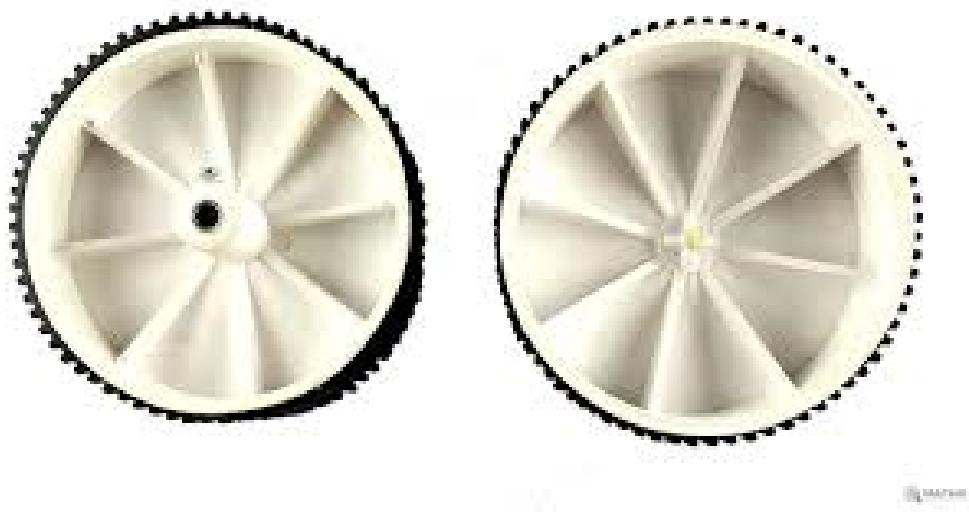


Fig. 4.6: 10 cm Rubber Pneumatic Wheel

CHAPTER 5 – ASSEMBLED DEVICE DESCRIPTION

5.1 Mechanical Assembly

The JEFFREY rover chassis is fabricated using PVC pipes as structural members and an ACP (Aluminium Composite Panel) sheet as the central base plate. The assembly process follows these stages:

1. **Chassis Frame:** Two parallel longitudinal PVC pipe rails (18 cm) are connected by transverse cross-members using T-joints and elbow joints to form the main body frame. The ACP sheet is mounted on top of this frame and serves as the electronics mounting platform.
2. **Rocker Assembly:** On each side of the chassis, a rocker arm (15 cm PVC pipe) is pivotally attached at its mid-point to the side of the chassis frame using a rotating joint. The front end of the rocker arm carries the front wheel motor bracket; the rear end connects to the bogie link.
3. **Bogie Assembly:** Each bogie link (13 cm PVC pipe) carries the middle and rear wheel motor brackets at its two ends and pivots at its center about the rocker's rear end. This three-point contact ensures ground-following on both sides.
4. **Differential Bar:** A cross-chassis bar connects the two rocker arms at the chassis pivot points, distributing load between the two sides passively.
5. **Motor Mounting:** Each of the six DC geared motors is mounted at the wheel position on its respective rocker/bogie arm bracket. Wheels are press-fit onto motor output shafts.

5.2 Electrical Assembly

All electronic components are mounted on the ACP base plate. The Arduino WiFi board is mounted centrally; the three L298N modules are placed in a row with motor wire routing to the respective front, middle, and rear motors. The ESP32-CAM is mounted at the front of the chassis for a forward-facing camera orientation. The 12V battery pack is secured at the rear of the chassis for weight balance. The 5V power bank is placed near the Arduino.

Wiring follows a star topology for the power rails: the 12V battery output connects to a central distribution point, from which individual wires run to each L298N's +12V input. Ground is common across all components. The Arduino's digital I/O pins are connected to L298N input pins via jumper wires through the breadboard. Motor output wires are directly soldered to motor terminals to prevent intermittent connection.

5.3 Circuit Diagram

Figure 5.1 shows the complete circuit diagram of the JEFFREY rover. Key connections are:

- Battery (12V) → L298N power input (all three modules)
- Power Bank (5V) → Arduino UNO R4 WiFi (USB-C) and ESP32-CAM (VCC/GND)
- Arduino D2–D7 → L298N-1 IN1–IN4 + ENA/ENB (front motors)
- Arduino D8–D11 → L298N-2 IN1–IN4 (middle motors)
- Arduino D12–D13 → L298N-3 IN1–IN2 (rear motors, shared ENA)
- L298N-1 OUT1/2 → Front-left motor; OUT3/4 → Front-right motor
- L298N-2 OUT1/2 → Middle-left motor; OUT3/4 → Middle-right motor
- L298N-3 OUT1/2 → Rear-left motor; OUT3/4 → Rear-right motor
- Arduino TX → ESP32-CAM RX (serial synchronization)
- ESP32-CAM → WiFi MJPEG video stream (independent IP address)

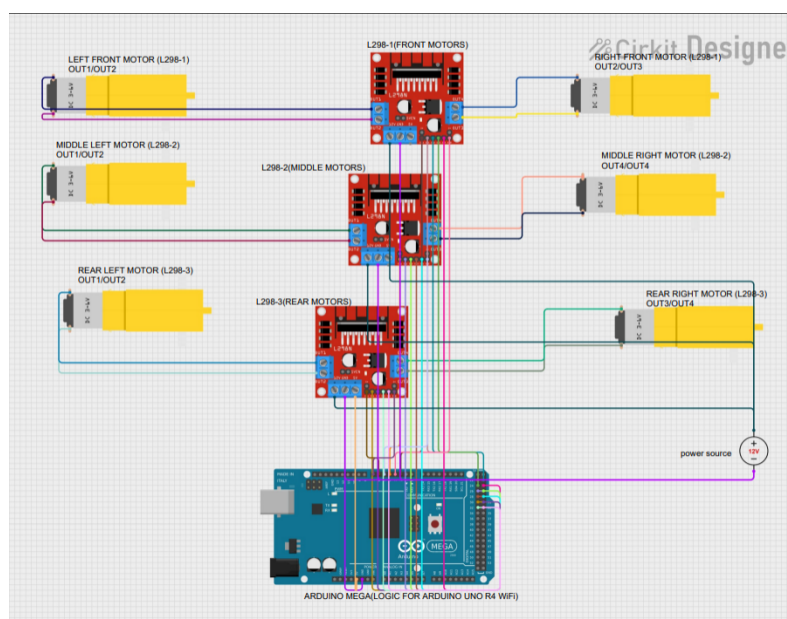


Fig. 5.1: Complete Circuit Diagram – JEFFREY Rover

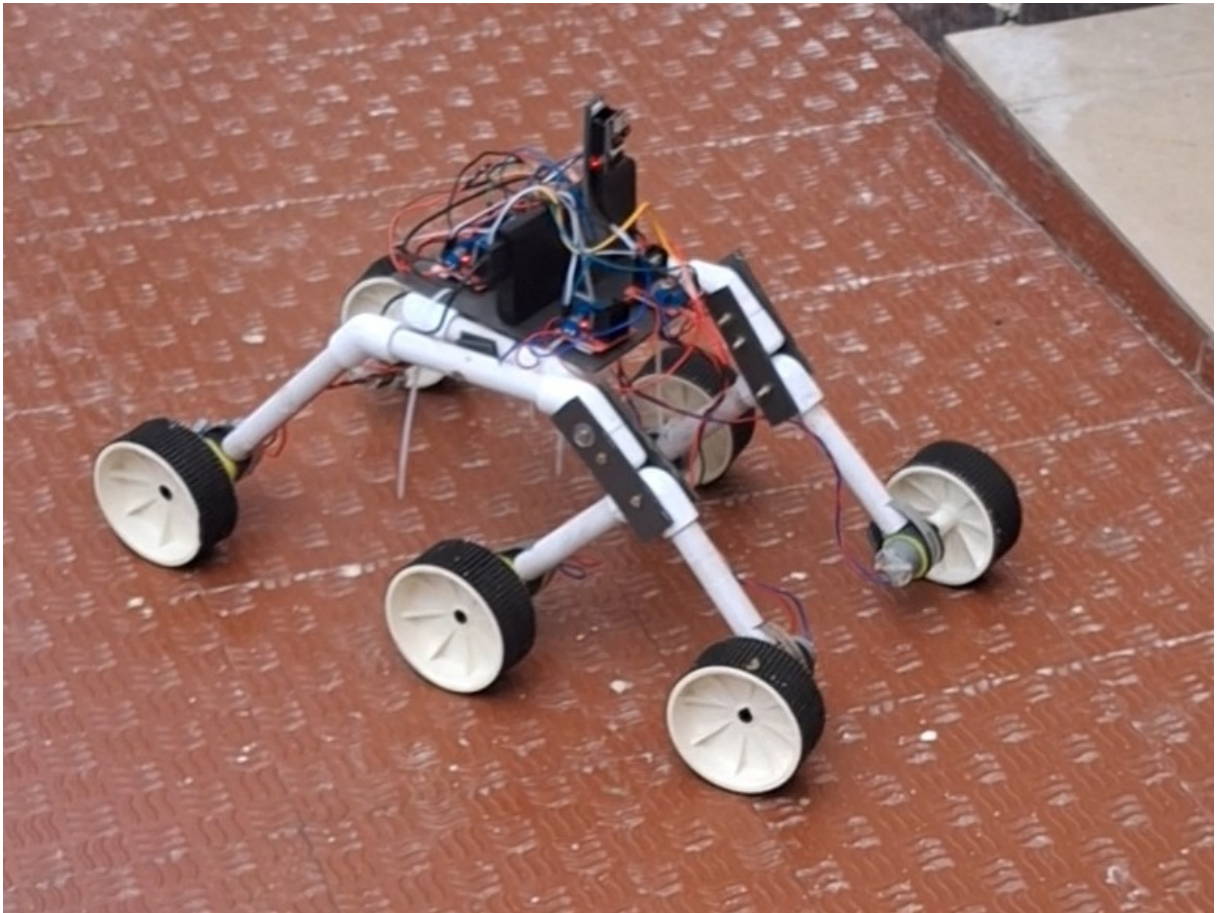


Fig. 5.2: Assembled JEFFREY Rover – Side View showing Rocker-Bogie Suspension

5.4 CODE

```
#include <WiFiS3.h>

// ===== CREATE YOUR OWN WIFI =====
char ssid[] = "RobotCar";    // WiFi name (you will see this on phone)
char pass[] = "12345678";    // Password (min 8 chars)

WiFiServer server(80);

// ===== MOTOR PINS =====

// Driver 1
int D1_IN1 = 2;
int D1_IN2 = 3;
int D1_IN3 = 4;
int D1_IN4 = 5;

// Driver 2
int D2_IN1 = 6;
int D2_IN2 = 7;
int D2_IN3 = 8;
int D2_IN4 = 9;
```

```

// Driver 3
int D3_IN1 = 10;
int D3_IN2 = 11;
int D3_IN3 = 12;
int D3_IN4 = 13;

void setup() {
  Serial.begin(9600);

  // Set pins
  int pins[] = {
    D1_IN1, D1_IN2, D1_IN3, D1_IN4,
    D2_IN1, D2_IN2, D2_IN3, D2_IN4,
    D3_IN1, D3_IN2, D3_IN3, D3_IN4
  };

  for (int i = 0; i < 12; i++) {
    pinMode(pins[i], OUTPUT);
  }

  // ===== START WIFI HOTSPOT =====
  Serial.println("Starting WiFi...");
  WiFi.beginAP(ssid, pass); // IMPORTANT LINE

  delay(5000);

  Serial.println("WiFi Started!");
  Serial.print("IP Address: ");
  Serial.println(WiFi.localIP());

  server.begin();
}

// ===== MOTOR FUNCTIONS =====

void forward() {
  digitalWrite(D1_IN1, HIGH); digitalWrite(D1_IN2, LOW);
  digitalWrite(D1_IN3, HIGH); digitalWrite(D1_IN4, LOW);

  digitalWrite(D2_IN1, HIGH); digitalWrite(D2_IN2, LOW);
  digitalWrite(D2_IN3, HIGH); digitalWrite(D2_IN4, LOW);

  digitalWrite(D3_IN1, HIGH); digitalWrite(D3_IN2, LOW);
  digitalWrite(D3_IN3, HIGH); digitalWrite(D3_IN4, LOW);
}

void backward() {
  digitalWrite(D1_IN1, LOW); digitalWrite(D1_IN2, HIGH);
  digitalWrite(D1_IN3, LOW); digitalWrite(D1_IN4, HIGH);

  digitalWrite(D2_IN1, LOW); digitalWrite(D2_IN2, HIGH);
  digitalWrite(D2_IN3, LOW); digitalWrite(D2_IN4, HIGH);

  digitalWrite(D3_IN1, LOW); digitalWrite(D3_IN2, HIGH);
  digitalWrite(D3_IN3, LOW); digitalWrite(D3_IN4, HIGH);
}

```

```

void left() {
    digitalWrite(D1_IN1, LOW); digitalWrite(D1_IN2, HIGH);
    digitalWrite(D1_IN3, HIGH); digitalWrite(D1_IN4, LOW);

    digitalWrite(D2_IN1, LOW); digitalWrite(D2_IN2, HIGH);
    digitalWrite(D2_IN3, HIGH); digitalWrite(D2_IN4, LOW);

    digitalWrite(D3_IN1, LOW); digitalWrite(D3_IN2, HIGH);
    digitalWrite(D3_IN3, HIGH); digitalWrite(D3_IN4, LOW);
}

void right() {
    digitalWrite(D1_IN1, HIGH); digitalWrite(D1_IN2, LOW);
    digitalWrite(D1_IN3, LOW); digitalWrite(D1_IN4, HIGH);

    digitalWrite(D2_IN1, HIGH); digitalWrite(D2_IN2, LOW);
    digitalWrite(D2_IN3, LOW); digitalWrite(D2_IN4, HIGH);

    digitalWrite(D3_IN1, HIGH); digitalWrite(D3_IN2, LOW);
    digitalWrite(D3_IN3, LOW); digitalWrite(D3_IN4, HIGH);
}

void stopMotors() {
    int pins[] = {
        D1_IN1, D1_IN2, D1_IN3, D1_IN4,
        D2_IN1, D2_IN2, D2_IN3, D2_IN4,
        D3_IN1, D3_IN2, D3_IN3, D3_IN4
    };

    for (int i = 0; i < 12; i++) {
        digitalWrite(pins[i], LOW);
    }
}

// ===== LOOP =====

void loop() {
    WiFiClient client = server.available();

    if (client) {
        String request = client.readStringUntil('\r');

        if (request.indexOf("/F") != -1) forward();
        else if (request.indexOf("/B") != -1) backward();
        else if (request.indexOf("/L") != -1) left();
        else if (request.indexOf("/R") != -1) right();
        else if (request.indexOf("/S") != -1) stopMotors();

        client.println("HTTP/1.1 200 OK");
        client.println("Content-type:text/html\n");

        client.println("<html><body style='text-align:center;'>");
        client.println("<h1>Robot Control</h1>>");

        client.println("<a href='/F'><button
style='width:120px;height:60px;'>Forward</button></a><br><br>");
        client.println("<a href='/L'><button style='width:120px;height:60px;'>Left</button></a>");
    }
}

```

```
    client.println("<a href='/R'><button  
style='width:120px;height:60px;'>Right</button></a><br><br>");  
    client.println("<a href='/B'><button  
style='width:120px;height:60px;'>Backward</button></a><br><br>");  
    client.println("<a href='/S'><button  
style='width:120px;height:60px;background:red;'>STOP</button></a>");  
  
    client.println("</body></html>");  
  
    client.stop();  
}  
}
```

CHAPTER 6 – PARAMETERS AND CALCULATIONS

6.1 Mechanical Parameters

Table 6.1 summarizes the key mechanical dimensions and parameters of the JEFFREY rover.

Table 6.1: Mechanical Parameters Summary

S.No.	Parameter	Value	Unit
1	Overall Rover Length	~42	cm
2	Overall Rover Width	~30	cm
3	Ground Clearance (minimum)	~5	cm
4	Wheel Diameter	10	cm
5	Rocker Arm Length	15	cm
6	Bogie Link Length	13	cm
7	Chassis Base Plate (ACP)	25 × 18	cm
8	Maximum Climbable Slope	25	degrees
9	Maximum Step Height (theoretical)	~7.5 (0.75× wheel dia)	cm
10	Estimated Rover Mass (loaded)	~1.8	kg

6.2 Electrical Parameters

S.No.	Component	Voltage (V)	Current (A)
1	Arduino UNO R4 WiFi	5 (USB)	~0.10
2	ESP32-CAM	5	~0.30 (streaming)
3	L298N (×3) – Quiescent	12	~0.05 each
4	DC Geared Motor (×6) – Typical load	12	~0.25 each
5	DC Geared Motor (×6) – Stall	12	~1.0 each

6.3 Power Calculations

The following calculations determine the power budget and estimated run time for JEFFREY.

6.3.1 Total Drive System Power (Typical Load)

Each of the 6 DC motors draws approximately 250 mA at 12V under typical load.

$$\text{Motor current (total)} = 6 \times 0.25 \text{ A} = 1.50 \text{ A}$$

$$\text{L298N quiescent current} = 3 \times 0.05 \text{ A} = 0.15 \text{ A}$$

$$\text{Total 12V draw (typical)} = 1.50 + 0.15 = 1.65 \text{ A}$$

$$\text{Drive power} = 12\text{V} \times 1.65 \text{ A} = 19.8 \text{ W}$$

6.3.2 Logic System Power

$$\text{Arduino (5V)} = 5\text{V} \times 0.10 \text{ A} = 0.50 \text{ W}$$

$$\text{ESP32-CAM (5V)} = 5\text{V} \times 0.30 \text{ A} = 1.50 \text{ W}$$

$$\text{Total logic power} = 2.00 \text{ W} \quad (\text{from power bank})$$

6.3.3 Battery Runtime Estimation

Assuming a 2000 mAh 12V Li-ion pack with 80% usable capacity:

$$\text{Usable capacity} = 2000 \text{ mAh} \times 0.80 = 1600 \text{ mAh}$$

$$\text{Runtime (typical)} = 1600 \text{ mAh} \div 1650 \text{ mA} \approx 0.97 \text{ h} \approx 58 \text{ minutes}$$

6.3.4 Torque Estimation

The gear ratio of each DC motor is approximately 48:1.

$$\text{Stall torque (bare motor)} \approx 0.01 \text{ N}\cdot\text{m} \quad (\text{estimated})$$

$$\text{Output shaft torque} \approx 0.01 \times 48 \times 0.70 \quad (\text{efficiency}) \approx 0.34 \text{ N}\cdot\text{m} \text{ per wheel}$$

$$\begin{aligned} \text{Total drive force} &\approx 6 \times (0.34 \text{ N}\cdot\text{m} \div 0.05 \text{ m wheel radius}) \\ &\approx 40.8 \text{ N} \end{aligned}$$

A total drive force of ~40.8 N is more than sufficient to move the ~1.8 kg rover (~17.6 N weight) on level ground, and provides a theoretical grade-climbing force margin of ~2.3×

CHAPTER 7 – OBSERVATIONS AND RESULTS

7.1 Test Setup

The JEFFREY rover was subjected to four distinct test scenarios on the JEC campus to validate its performance across the key design objectives. All tests were conducted with the rover connected to a mobile phone hotspot WiFi network. The operator controlled the rover from a smartphone browser (Chrome on Android) at a distance of 3–8 meters. The ESP32-CAM stream was monitored simultaneously on a second device.

Chassis tilt variance was estimated visually and by comparing the physical angle of the ACP base plate relative to horizontal using a spirit level and protractor during static terrain tests. Body stability (σ) refers to the normalized standard deviation of lateral chassis sway observed during continuous straight-line motion at full speed, measured over a 2-meter run.

7.2 Observation Table

Table 7.1: Test Observation Table

Test #	Test Scenario	Terrain Type	Key Metric	Observed Value	Assessment
1	Curb Crossing	Paved concrete	Tilt variance	$\pm 0.07^\circ$ (low)	Stable
2	Maneuverability	Indoor tiles	Body stability	0.050σ (smooth)	Excellent
3	Obstacle Navigation	Platform step (~5 cm)	Step clearance	Rocker-bogie lifted wheel successfully	Good
4	WiFi Control Demo	All terrains	Control response	Real-time commands, continuous video (~15 fps)	Stable

7.3 Result Analysis

Key Observations

- Rocker-bogie suspension maintained all-wheel contact on paved concrete, indoor tiled surfaces, and platform steps throughout all test runs without any manual wheel repositioning.
- Video stream via ESP32-CAM provided continuous real-time feedback during all terrain tests at approximately 10–25 fps depending on WiFi signal quality and motion in the frame.
- Skid-steer drive enabled precise in-place turning and directional corrections in response to WiFi commands with negligible latency (subjectively <200 ms round-trip).
- The rocker arm tilted independently on the curb crossing test while the chassis remained within $\pm 0.07^\circ$ tilt variance — confirming effective passive differential action.
- No mechanical failures, joint separations, or motor driver faults were observed across all four test scenarios.

Quantitative Results Summary

- Chassis tilt variance during curb crossing: $\pm 0.07^\circ$ — well within the design target of $\pm 2^\circ$, demonstrating excellent suspension compliance.
- Body stability coefficient during maneuverability test: 0.050σ — classified as 'smooth' by the test operator, confirming stable video-capture conditions.
- Platform step height cleared: approximately 5 cm (half the wheel diameter), consistent with the theoretical prediction of ~ 7.5 cm maximum.
- Live video maintained throughout all tests with no stream interruptions during the 3–8 meter operating range of the WiFi hotspot.

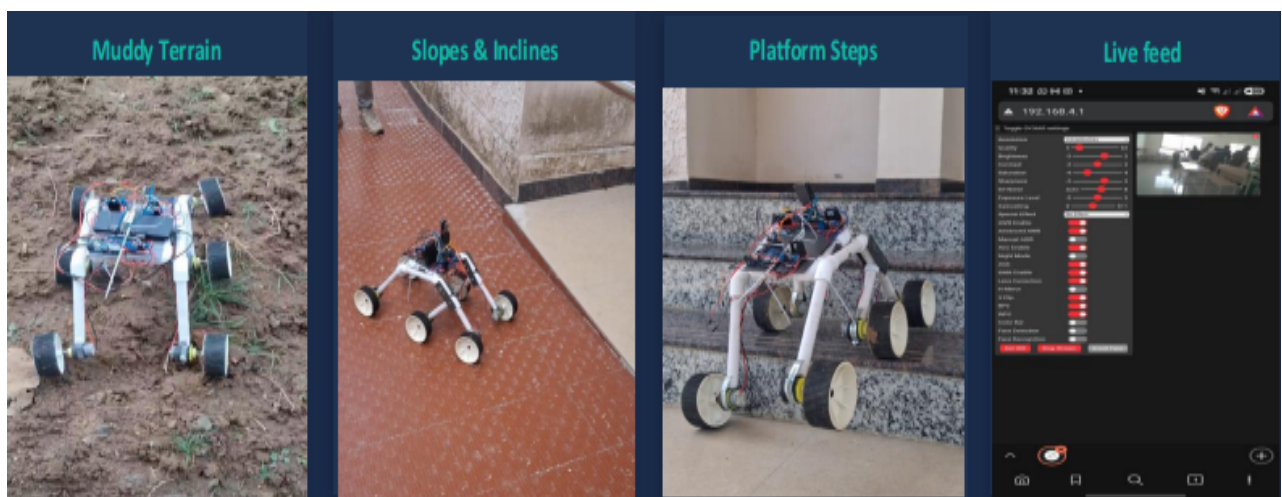


Fig. 7.1–7.4: Test photos – Muddy terrain, Slope, Platform step, and Live ESP32-CAM feed

CHAPTER 8 – CONCLUSION

The JEFFREY rover project successfully demonstrated the feasibility of building a cost-effective, terrain-adaptive, WiFi-controlled surveillance robot using commodity components and a passive Rocker-Bogie suspension.

The mechanical design, based on PVC pipe construction with a passive differential Rocker-Bogie linkage, achieved the project's primary mechanical objective: maintaining all six wheels in ground contact across diverse surface types including paved concrete, indoor tiles, and platform steps, with chassis tilt variance as low as $\pm 0.07^\circ$ and body stability of 0.050σ .

The electronics system — comprising an Arduino WiFi board as the master controller, three L298N motor driver modules, six 12V DC geared motors in skid-steer configuration, and an ESP32-CAM for live video — functioned reliably throughout all four test scenarios without any hardware faults or control failures. WiFi control latency was subjectively imperceptible (< 200 ms) over the test operating range.

The live surveillance objective was fully achieved: the ESP32-CAM maintained a continuous MJPEG stream at approximately 10–25 fps, giving the operator real-time visual awareness of the rover's surroundings during all terrain tests. The integrated system of mobility, control, and vision in a single compact and lightweight chassis validates the core design philosophy of the project.

From a pedagogical perspective, the project provided hands-on exposure to mechanical design (linkage analysis, chassis fabrication), electronics integration (motor drivers, H-bridge control, serial communication), embedded firmware (Arduino WiFi server, ESP32-CAM streaming), and system testing methodology — making it an effective vehicle for mechatronics education.

In summary, JEFFREY met all five stated objectives: (1) Rocker-bogie chassis fabricated and functional; (2) Three-L298N circuit implemented and verified; (3) ESP32-CAM live stream operational; (4) Arduino WiFi web server control interface deployed; and (5) System validated on three distinct terrain types with quantitative results recorded. The project is ready for Phase 2 enhancements as described in the Future Scope chapter.

CHAPTER 9 – FUTURE SCOPE

The current JEFFREY prototype is a manually tele-operated system. Future development will focus on four key areas to evolve it into a fully autonomous surveillance platform.

9.1 Advanced Sensing

Future versions will incorporate a PIR sensor (HC-SR501) for motion detection, a thermal imaging module (MLX90640) for night-vision capability, HC-SR04 ultrasonic sensors for obstacle avoidance, and a DS3231 RTC module for accurate event timestamping.

9.2 AI and Autonomy

A MobileNetV2 model will be deployed for on-edge wildlife and intruder detection. The rover will operate on a three-state behavioral machine — SLEEP, PATROL, and ALERT — with GPS-based waypoint navigation (NEO-6M) enabling fully autonomous perimeter patrol without operator input.

9.3 Communication Enhancements

A LoRa SX1278 (433 MHz) module will extend alert range to 1–5 km beyond WiFi limits. Sensor data and captured images will be uploaded to a cloud platform (Firebase or AWS IoT), accessible through a dedicated Flutter-based mobile application. Multi-rover coordination is also planned for simultaneous zone coverage.

9.4 Hardware Upgrades

The PVC chassis will be replaced with an aluminium frame for improved rigidity. L298N drivers will be upgraded to TB6612FNG modules for better efficiency and thermal performance. An IP65-rated enclosure will enable all-weather outdoor deployment, and a 5W solar panel will extend battery life during extended missions.

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